Budget Anchored Region Reduction for Collaborative Responsibility Repair

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***Abstract*—Localized cost changes can require responsibility transfers under a hard handoff budget. We derive Budget Anchored Region Reduction (BARR) for binary assignments on a fixed coordination graph. Perturbation support, positive node-wise handoff costs, and an old-plan certificate jointly define a safe region. Its restricted optimum preserves the full budgeted value when the old plan is optimal, and otherwise loses at most the certified old-objective gap. Reverting all unanchored modification components together incurs this gap once. Boundary absorption transfers any restricted solver's bounds to the full problem. Matched CPU experiments include weighted-region ablation, graphs up to 2,048 vertices, and semi-synthetic ownership repair on publicly sourced CAD co-edit graphs. A rational graph-cut Lagrangian baseline compares time, feasible quality, and certificates. Results support conditional preprocessing benefits, not universal solver superiority: many public cases terminate by one cut, dense graphs gain little, and retained hard cores can remain unresolved.**

***Index Terms*—collaborative design, responsibility allocation, budgeted repair, region reduction, optimality certificate**

I. Introduction

Responsibility allocation in collaborative design often starts from an existing plan. A requirement update may increase execution costs for a few work packages, while transferring responsibilities consumes documentation, onboarding, or interface handoff effort. Recomputing an unconstrained assignment can therefore produce an operationally inadmissible change. A hard upper bound on handoff effort makes this restriction explicit.

We study a restricted responsibility layer: each work package can be assigned to either of two teams already qualified to execute it. Unary terms represent execution costs, and attractive pairwise terms charge for interfaces crossing team boundaries. A repair changes responsibilities within a weighted budget. The model does not choose start times or enforce workload capacities. This layer is relevant only after qualification and capacity feasibility have been handled: for example, two approved maintenance pools with reserved effort and no active workload cap on the selected changes. This is a deployment condition, not an inference from a dependency graph. Active capacities can couple distant modifications and invalidate our argument.

An unconstrained binary attractive energy is minimized by a graph cut [1]. Adding a weighted change constraint leaves a discrete optimization problem for which a returned unconstrained optimum may be infeasible. Must every vertex then be included in the constrained model? Local cost changes suggest a smaller region, but distance from changed vertices alone does not justify fixing distant labels. A pre-existing suboptimal plan may have valuable repairs anywhere in the graph.

Our contribution is a structural reduction. Changed unary factors define anchors. The old plan's objective gap bounds the cost of reverting all modification components that avoid these anchors. Remaining modifications have paths to anchors whose total handoff cost cannot exceed the budget. This yields a computable activity region and

$$\mathrm{OPT}_B \le \mathrm{OPT}_K \le \mathrm{OPT}_B + \varepsilon, \tag{1}$$

where OPT_B is the unrestricted budgeted optimum and OPT_K fixes the exterior to the old plan. For $\varepsilon = 0$, the reduction preserves the optimal value. It preserves the existence of a suitable optimum; it need not preserve every tied optimum.

Public design records motivate such an ownership layer: Gopsill et al. document shared CAD-file work in three Formula Student vehicle projects [2], [3]. We derive co-edit association graphs from their open logs, then impose synthetic binary ownership and handoff costs. The source's A/B/C labels identify three projects, not our two responsibility pools. The data establish a design-work structure, but do not establish team interchangeability, spare capacity, or measured transfer savings.

We evaluate BARR before an unchanged CP-SAT backend, with matched cut shortcuts and charged preprocessing. Weighted-region ablation isolates its geometry; inexact plans test the certificate. A separate Lagrangian-cut route exposes quality–time tradeoffs. Supplementary materials contain code, source attribution, fixed inputs, all saved outcomes, and file hashes.

II. Related Work

CSCWD allocation topics include personnel sharing [4], cooperation/conflict [5], and qualified, capacity-limited divisible workloads [6]. Our model addresses a narrower responsibility layer; it does not enforce or outperform those different resource models.

Table I locates the proposed inference. Hamming neighborhoods, limited refinement, graph cuts, Lagrangian bounds, and safe variable fixing are established tools. Our contribution is specifically the safe exterior-fixing rule jointly obtained

from perturbation support, weighted recourse, and an old-objective gap. The table compares the cited formulations; their extensions may require separate analysis.

Dynamic cuts reuse residual flows for unconstrained updates [12]; stable dynamic cuts concern vertex arrival and flip stability [13]. They do not directly supply the stated fixed-graph budget certificate. Nor can unconstrained persistency simply be reused: with isolated vertices, $x^0 = (0, 0)$, $B = 1$, and new unary pairs $(0, -2), (0, -1)$, fixing both unconstrained-optimal labels to $(1, 1)$ removes every feasible repair. Section IV-E shows why the old certificate is also necessary. Neither counterexample rules out deriving our result from some other prior theorem.

III. Problem Formulation

Let $G = (V, E)$ be fixed and undirected, with $n = |V|$ and $m = |E|$. Vertex i is a work package and $x_i \in \{0, 1\}$ selects one of two prequalified teams. For $t \in \{0, 1\}$, define

$$E_t(x) = \sum_{i \in V} c_i^t(x_i) + \sum_{ij \in E} w_{ij} \mathbf{1}[x_i \neq x_j], \tag{2}$$

with integer unary costs and fixed integer $w_{ij} \geq 0$. Given old assignment x^0, handoff effort is

$$R(x, x^0) = \sum_{i \in V} r_i \mathbf{1}[x_i \neq x_i^0], \qquad r_i \in \mathbb{Z}_{>0}. \tag{3}$$

For integer $B \geq 0$, the repair is

$$\mathrm{OPT}_B = \min\{E_1(x) : x \in \{0, 1\}^V,\ R(x, x^0) \leq B\}. \tag{4}$$

The old plan is feasible; there are no additional hard constraints. Execution costs c_i^1 affect the objective, whereas handoff costs r_i consume the change budget.

Only unary factors on $A = \{i : c_i^0 \neq c_i^1\}$ may change. Both positive and negative shifts are allowed. We require a certified $\varepsilon \geq 0$ satisfying

$$E_0(x^0) \leq \min_x E_0(x) + \varepsilon. \tag{5}$$

Our implementation computes the old unconstrained minimum exactly by graph cut, making ε the actual old-objective gap. A valid old lower bound can instead supply a conservative certificate. Optimality cannot be inferred from an organization's use of the old plan. Negative unary entries are handled by retaining label-independent offsets when constructing nonnegative terminal capacities.

IV. Budget Anchored Region Reduction

A. Reverting Remote Modification Components

For a budget-feasible x, let $S = \{i : x_i \neq x_i^0\}$. A connected component of the induced graph $G[S]$ is anchored if it intersects A.

Lemma 1 (Union reversion): Let T be the union of all components of $G[S]$ disjoint from A. Restore x_i^0 on T and retain x_i elsewhere, obtaining y. Then $R(y, x^0) \leq R(x, x^0)$ and $E_1(y) \leq E_1(x) + \varepsilon$.

Proof: Let z equal x on T and x^0 elsewhere. No edge joins T to $S \setminus T$, since these sets contain different components of $G[S]$. Every neighbor outside S keeps its old label in both x and y. Moreover, $c_i^1 = c_i^0$ on T. Expanding unary and crossing-edge differences gives

$$E_1(x) - E_1(y) = E_0(z) - E_0(x^0) \geq -\varepsilon. \tag{6}$$

The inequality follows from (5). Restoring vertices removes positive resource terms. Applying the old-gap bound once to the union T incurs one ε, rather than one per component. ∎

B. A Budget-Defined Activity Region

Charge every vertex on a path $P : a \rightsquigarrow v$, including both endpoints. Define

$$d_r(A, v) = \min_{a \in A,\, P : a \rightsquigarrow v} \sum_{i \in P} r_i, \quad K_B = \{v : d_r(A, v) \leq B\}. \tag{7}$$

An unreachable vertex has infinite distance; $A = \varnothing$ gives $K_B = \varnothing$. Positive costs bound the path length of feasible changed components. An anchor with $r_a > B$ cannot itself be changed.

Theorem 1 (Region reduction): Under Section III's assumptions, let OPT_K be the optimum of (4) with $x_i = x_i^0$ outside K_B. Then (1) holds. If $\varepsilon = 0$, at least one global budgeted optimum is supported in K_B.

Proof: Take an optimum x^* of (4) and apply Lemma 1 to obtain y. Every remaining changed component intersects A. For a changed v, a simple path to an anchor lies inside its component. Every path vertex is changed, so its node-cost sum is at most $R(y, x^0) \leq B$. Thus $v \in K_B$, and y is feasible for the restricted problem. Hence $\mathrm{OPT}_K \leq E_1(y) \leq \mathrm{OPT}_B + \varepsilon$. Restricting the feasible set gives $\mathrm{OPT}_B \leq \mathrm{OPT}_K$. For $\varepsilon = 0$, the values coincide. ∎

For unit costs, K_B is the at-most-$(B-1)$-hop neighborhood of A when $B \geq 1$; $K_0 = \varnothing$. Charging the anchor is essential. Letting $r_{\min} = \min_i r_i$, define the coarse region

$$H_B = \{v : 1 + \mathrm{dist}_G(A, v) \leq \lfloor B / r_{\min} \rfloor\}. \tag{8}$$

A path's cost is at least $r_{\min}$ times its number of vertices, implying $K_B \subseteq H_B$. Fixing the exterior of H_B is also safe, but can retain unnecessary vertices under heterogeneous costs. We use it as a mechanism ablation, not a separate contribution.

C. Boundary Absorption and Certificates

For each crossing edge ij, $i \in K = K_B$, $j \notin K$, absorb its interaction into the retained unary:

$$\widetilde{c}_i(a) = c_i^1(a) + \sum_{j \notin K :\, ij \in E} w_{ij} \mathbf{1}[a \neq x_j^0]. \tag{9}$$

Let C sum exterior unaries at x^0 and edges entirely outside K. Retaining internal edges yields

$$E_1(x) = E_K(x_K) + C \tag{10}$$

for every reconstruction. Dropping crossing edges would solve a different problem. Only retained vertices consume the restricted change budget.

TABLE I
CLOSEST FORMULATIONS AND THE ADDITIONAL INFERENCE STUDIED HERE. A MISSING ASSUMPTION MEANS IT IS NOT REQUIRED BY THE CITED FORMULATION; IT IS NOT A CLAIM OF IMPOSSIBILITY.

Work	Perturbation setting	Modification control	Old-plan requirement	Guarantee or objective
Local branching [7]	General binary model	Hamming neighborhood	Feasible incumbent	Explore an incumbent neighborhood by integer optimization
Binary-energy trust region [8]	Binary pairwise energy	Hamming bound / dual	Current iterate	Graph-cut Lagrangian bound in energy approximation
Submodular reassignment [9]	Multiple task scenarios	Distances to minimizer sets	First-stage plan is a variable	Optimize scenario-wise reassignment distances
OptiRefine [10]	Densest-subgraph / maximum-cut objectives	Exactly k unit refinements	Given graph solution	Approximation, hardness, and refinement algorithms
Safe SFM screening [11]	Fixed submodular objective	No handoff-budget model	Proximal optimization certificate	Fix active/inactive elements for that minimization
BARR	Local unary changes on a fixed graph	Weighted upper budget B	Certified old-objective gap	Fix outside K_B; exact value at zero gap, additive loss at most the gap

Corollary 1 (Global interval): If a restricted solver returns $L_K \le \mathrm{OPT}_K \le U_K$, including C, its feasible solution reconstructs to a feasible full assignment and

$$L_K - \varepsilon \le \mathrm{OPT}_B \le U_K. \tag{11}$$

Proof: Reconstruction preserves resource usage and, by (10), energy. Feasibility supplies the upper bound, and Theorem 1 gives $\mathrm{OPT}_B \ge \mathrm{OPT}_K - \varepsilon \ge L_K - \varepsilon$. ∎

A restricted exact solve is globally exact when its transferred bounds close. With $\varepsilon > 0$, restricted optimality alone gives an additive certificate.

D. Algorithm and Cost

Both pipelines first solve the new unconstrained cut problem. If its returned assignment satisfies B, its value is also a feasible constrained lower bound, proving optimality. Both pipelines then terminate. Otherwise BARR computes the old certificate, extracts A, constructs K_B, absorbs boundary factors, and invokes the constrained backend. Multi-source Dijkstra initializes $d(a) = r_a$ and relaxes toward a neighbor j by adding r_j.

In order, the query performs the shared new cut, certifies the old gap if needed, computes the weighted region, absorbs its boundary, solves the retained budget model, and reconstructs a feasible full assignment with bounds $[L_K - \varepsilon, U_K]$.

Given a certificate, support extraction, region computation and reduced-instance construction cost $O((n+m)\log n)$ time and $O(n+m)$ memory. A cold query additionally pays for the new cut and, if needed, the old cut. Writing their costs as T_{cut} and constrained solving time as $T_{\mathrm{solve}}(K, B)$, a non-shortcut query costs

$$T_{\mathrm{cut}}(E_1) + T_{\mathrm{cut}}(E_0) + O((n+m)\log n) + T_{\mathrm{solve}}(K, B). \tag{12}$$

The backend can remain exponential; no polynomial-time budget algorithm or polynomial kernel follows. Low graph diameter can make $K_B = V$, leaving overhead without reducing the model.

Our CP-SAT formulation uses binary x_i and edge variables $y_{ij} = |x_i - x_j|$, objective (2), and one linear budget constraint. Full and reduced graphs use the same formulation, settings, and old-plan hints.

E. Why the Old Certificate Is Necessary

Take two disconnected vertices, $x^0 = (0,0)$, $r = (1,1)$, $B = 1$. Old unary pairs are $(0,5)$ and $(10,0)$; only the first changes to $(1,5)$. The old gap is 10. The region contains only vertex one; its restricted optimum has value 11. Changing remote vertex two instead gives full optimum 1. The loss equals ε, disproving exact locality for arbitrary old plans and showing that the bound can be tight. Hard capacities cause a separate failure: reverting a remote component can destroy compensating balance elsewhere, invalidating Lemma 1's feasibility step.

V. EXPERIMENTAL EVALUATION

A. Protocol and Earlier Mechanism Tests

Experiments use one arm64 MacBook Pro, Python 3.12.14, OR-Tools 9.15.6755, and one CP-SAT worker. FullCP and BARRCP share the formulation, old-plan hint, and exact new-cut shortcut. Queries include cuts, preprocessing, model and solver construction, and solving; BARR pays to recompute its old certificate. Common imports/warm-up and old-plan generation are outside timing. Targets are soft, not process deadlines. Inputs, configuration/source hashes, labels, bounds, and all failures/incomplete solves are retained.

The original main study crosses $n \in \{128, 512\}$, modular/grid/small-world graphs, edge coupling 2/12, one/four anchors, budgets 8/16, and ten seeds: 480 combinations. Modular graphs join 4×4 blocks by unit bridges; small-world graphs add $2n$ sampled pairs to a grid, merging duplicates. Nonbridge weights are uniform integers $[c, 2c]$. Unary base and preference penalties are uniform $[1, 4]$, with alternating preferred modules/rows. Old plans are exact cuts; anchor old-label costs increase by $[8c, 10c]$, while alternative costs decrease by at most one quarter of the increase. Handoff costs are one. No generated case is rejected.

After 96 calibration cases, the first seeds 100–102 were frozen; 103–109 later extended the same factors. Thus the entire study was not prospectively registered. Each method/case

TABLE II
ORIGINAL MAIN STUDY. SPEED IS PAIRED FULLCP/BARRCP; ALL-CASE STATISTICS RETAIN SHORTCUTS.

Family	n	CP/all	Speed CP	Speed all
Grid	128	69/80	1.00	1.00
Grid	512	54/80	1.88	1.37
Modular	128	29/80	1.30	1.00
Modular	512	30/80	3.17	1.01
Small-world	128	72/80	0.98	0.99
Small-world	512	64/80	0.99	0.99

TABLE III
WEIGHTED ABLATION. REGION COLUMNS: CP-REQUIRED MEDIAN PERCENTAGES. SPEED: HOPCP/BARRCP.

Family	CP/all	$H_B\%$	$K_B\%$	Speed CP	Speed all
Grid	56/80	56.7	20.0	2.59	2.13
Modular	40/80	22.0	9.8	1.56	1.03
Small-world	66/80	100.0	99.5	1.01	1.01

has three alternating-order timings, aggregated by its median before paired FullCP/BARRCP ratios. Intervals resample $(n, \text{family}, \text{seed})$ clusters 2,000 times, retaining other factors and repeats within each cluster. CP-required is determined by the shared shortcut, independently of timing.

All 2,880 main calls prove equal optima; 314/480 budgeted values strictly exceed the unconstrained minimum, and greedy is suboptimal on 55/480. Table II retains shortcuts and their frequency. A returned cut over budget alone would not prove budget activity because a tied optimum might be feasible. At 512 vertices, modular CP-required speed is 3.17, interval $[2.86, 3.86]$, but all-case speed is 1.01. Grid values are 1.88, $[1.54, 2.15]$, and 1.37. Small-world graphs gain little; most main cases are not hard searches.

A separate 240-combination, 512-vertex ablation uses r_i uniform in $\{1, 2, 3\}$ and seeds 200–209. FullCP, safe-hop HopCP, and BARRCP rotate through three timing positions; both regional methods pay for certification. All 2,160 calls prove the same global optima. Table III shows that weighted geometry helps grids and modules but scarcely reduces small-world graphs. Grid HopCP/BARRCP speed has interval $[2.41, 2.75]$, modular $[1.47, 1.68]$; repetitions are not independent cases.

For 32 inexact old plans on 128-vertex grids/modules, we flip one/three low-cost non-anchor vertices and recompute the complete old gap, including interactions. Gaps are 24–105 (median 58); 10 restricted optima are worse, with maximum loss 67, sometimes attaining ε. Full and restricted optima are proved separately; no speed comparison treats these as equal-quality outputs. Sixteen tests, including rational-cut bounds/primal recovery on 50 random enumerated graphs, pass. The earlier 1,200 small-instance enumeration checks 650 zero-gap and 550 positive-gap combinations without violating the reduction bound. Finite tests do not replace the proof.

B. Public Design Graphs and Model Mapping

The open Bath dataset [3] records CAD edits in three vehicle-design projects; its original study derives DSMs from co-edit frequencies [2]. Nodes in our application are CAD-file maintenance packages. A cross-pool edge cost is a proxy for coordination when associated files have different owners. Primary/backup qualification and reserved capacity must be established externally; the logs do not establish them. If workload balance is active, this experiment's ownership layer is insufficient and our feasibility proof does not apply.

We retain files with at least five distinct edit times, yielding 226/231/168 nodes in projects A/B/C. For each file edit at t, each other file edited before the earlier of the next same-file edit and $t + h$ counts once; $h = 1, 12$ hours. We average directional conditional frequencies, retain pairs at least $1/4$, and round eight times that average to integer weights. Every qualifying node, including isolates, remains. This is a disclosed co-edit proxy, not a physical interface or a reproduction of the complete original DSM pipeline. Edge counts are 2726/8152 (A), 535/693 (B), and 199/481 (C), for one/twelve hours. Figure 1 illustrates their contrasting density.

For each graph, subsystem order alternates synthetic pool preference, with nonpreferred costs uniform 4–8 and an exactly cut-optimal old plan. Handoff cost is one plus indicators for at least ten edits and assembly-file type. One/four uniformly sampled anchors increase old-label cost by 32–40 and decrease the alternative by 0–8; $B = 8, 16$. Three new seeds 900–902 yield 72 cases. These parameters were frozen before repair outcomes. Costs, qualifications, slack, and transfers are assumed, not observed. All 72 CP pairs prove equal optima, but 66 terminate by the shared shortcut: this is application-structure evidence, not evidence of substantial operational acceleration.

C. Lagrangian Route and Larger/Harder Inputs

LRRepair is our classical comparator implementation, not a reproduction of a complete named optimizer. It computes exact cuts of $E_1 + \lambda R$, with rational $\lambda \geq 0$ via integer scaling, and lower bounds $q(\lambda) - \lambda B$. After doubling brackets the feasible side, at most 24 bisections refine it. It retains the best feasible cut, restores the closest infeasible cut by reverting vertices of minimum energy increase per freed effort, and applies feasible single-flip descent from that candidate, the old plan, and the feasible cut. The dual phase gets at most 65% of the time target; recovery gets the remainder. The old plan remains a feasible fallback. Integer ceiling gives a valid energy lower bound. A finite multiplier search is not asserted to optimize the dual; primal recovery has no approximation guarantee. This provides a cut-based feasible route and a certificate, rather than comparing only generic exact backends.

We compare all three routes freshly on every previous 120 large and 20 core inputs, plus the 72 public cases (636 calls). The scale grid is $n = 1024, 2048$, three families, one/four anchors, $B = 16, 64$, coupling 12, perturbation scale 64, heterogeneous costs, and five seeds. It was frozen after 32 exploratory cases. Core instances contain a $q = 64, 128$ cycle

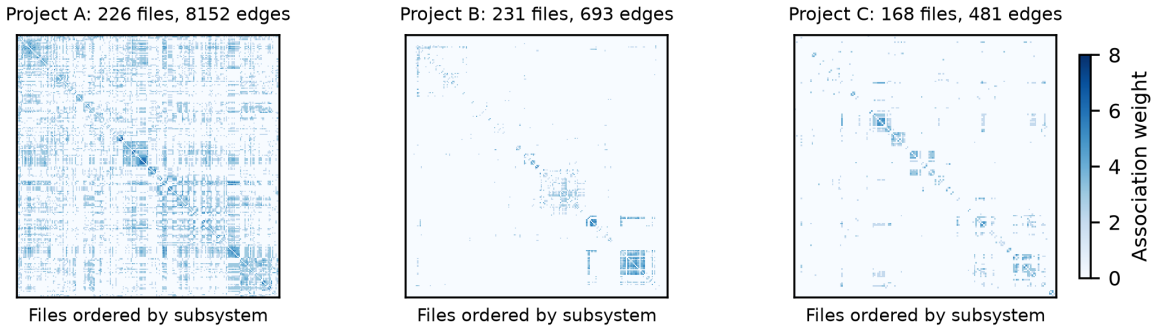


Fig. 1. Twelve-hour co-edit proxy matrices derived from public CAD logs [3], ordered by subsystem and file ID. Color is a quantized association weight, not measured handoff effort; binary ownership is synthetic. One-hour variants and isolated files are also retained.

plus random matching, unit internal edges, anchor-star edges of weight 3, and a dominated tail to reach n. Core unaries are $(0, -1)$, tail $(0, 10)$; anchor changes from $(0, 4q + 1)$ to $(0, -4q)$, with unit handoffs and $B = 1 + q/2$. The old all-zero plan is uniquely optimal; a new optimum changes the anchor and $q/2$ core vertices. Only q controls intrinsic search difficulty; tail growth tests exterior preprocessing.

Targets are three seconds for scale/public and five for cores, with rotating method order and one timing each. These descriptive comparisons reuse earlier inputs; they are not additional independent instances or pooled with triple-timing intervals. Table IV retains every censored call and reports quality alongside time.

On 2,048-vertex modular graphs, 14/20 cases require CP; their median paired FullCP/BARRCP speed is 7.30, versus 7.67 in the earlier timing stage. All-case median times are 0.148/0.016 s. For the 64-core family at that size, both CP routes certify 5/5; fresh paired speed is 3.24 (earlier 2.86). LRRepair is faster at 0.173 s median but has maximum actual loss 12 against proved optima and certifies none. On 128-core cases it can lose 26 on the eight cases with a proved reference optimum. The reduced CP route certifies 4/5 per size in this single run, versus 1/5 previously. These separate observations motivate the repeated-query check below; they do not establish a stable success probability or its cause.

LRRepair matches 71/72 public optima and certifies 69/72; its worst actual public loss is one. It also matches every proved grid optimum, though some dual bounds remain open. On four censored 2,048-vertex small-world cases, LRRepair gives a better feasible objective than both CP routes (up to 417 lower); its maximum returned gap is 8 versus 418 for CP. Small-world reduction remains ineffective. Thus cut recovery can be preferable for fast feasible repair, while BARR helps exact certification on suitable structure; neither route dominates time, quality, and bound closure. Complete per-case values and previous stages remain available.

D. Repeatability of Difficult Core Queries

We repeat all 20 saved core inputs ten times per CP route (400 calls), with the same five-second target, formulation, search settings and one worker. The frozen protocol fixes the original solver seed, shuffles case order by round, and alternates method order. Preliminary attempts exposed suspend-related timing discrepancies; all diagnostic calls are archived separately. We use macOS continuous monotonic time for accounting/deadlines, keep the laptop awake, and abort on a clock mismatch or pause delta above 0.1 s. The run is completed in two execution segments after a tool interruption; saved queries are not rerun. All 400 logged calls pass these checks. We launched no concurrent benchmark or document build.

Table V retains every incomplete query. The archive supplies each instance's ten outcomes, residual gaps, and timing components. Small time dispersion at the cap reflects censoring, not reliable time-to-optimum. Background load and power mode are not controlled. Fixed-seed repeats on one laptop characterize these two execution segments only; they do not establish cross-session or hardware reliability. Bound closure is counted from each call's own returned interval, even when another call establishes the optimum. We report no speed ratio for unresolved cores and no independent-trial confidence interval.

VI. Discussion and Limitations

The contribution is a value-preserving/additively certified preprocessing rule, not a claim that every optimum is local or that constrained search becomes polynomial. Dense connectivity can retain the full graph. Repeated events require recertification: a previous budgeted optimum need not be the next old unconstrained optimum.

Public co-edit structures improve traceability of the design setting but do not validate binary qualification, slack, coordination-cost calibration, or human efficiency. Active capacity, scheduling, and multi-team constraints require a new feasibility argument. There is no industrial transfer trial or peak-memory benchmark. The three-route comparison has single timings; the repeated-core stage tests repeat variability in two execution segments on the same machine. Neither stage establishes cross-machine stability. Previous runs are retained separately. Tighter old-plan certificates and feasible sets closed under union reversion are useful directions for extension.

TABLE IV

FRESH THREE-ROUTE COMPARISON. F/R/L = FULLCP/BARRCP/LRREPAIR; TIMES ARE ALL-CASE MEDIAN SECONDS. CERT COUNTS EACH METHOD'S CLOSED BOUNDS; GAP IS ITS MAXIMUM RETURNED ABSOLUTE BOUND GAP, RETAINING ALL INCOMPLETE CALLS. K IS THE CP-REQUIRED MEDIAN REGION PERCENTAGE; – MEANS EVERY CASE USES THE SHORTCUT. FASTER FEASIBLE OUTCOMES NEED NOT HAVE EQUAL QUALITY.

Input	n	CP/all	$K\%$	Cert F/R/L	Time F/R/L (s)	Max gap F/R/L
Grid	1024	20/20	70.5	20/20/11	0.189/0.115/0.081	0/0/7
Grid	2048	16/20	50.4	20/20/15	0.289/0.067/0.187	0/0/3
Modular	1024	15/20	13.7	20/20/6	0.075/0.012/0.059	0/0/26
Modular	2048	14/20	7.1	20/20/6	0.148/0.016/0.126	0/0/24
Small-world	1024	20/20	100.0	20/20/6	0.256/0.268/0.143	0/0/17
Small-world	2048	20/20	100.0	16/16/10	0.639/0.627/0.320	418/418/8
Core 64	1024	5/5	9.4	5/5/0	0.440/0.215/0.082	0/0/21
Core 64	2048	5/5	4.7	5/5/0	0.641/0.221/0.173	0/0/21
Core 128	1024	5/5	18.8	2/4/0	5.002/3.548/0.120	15/17/43
Core 128	2048	5/5	9.4	1/4/0	5.004/3.756/0.235	20/17/43
CAD A, 1 h	226	0/12	–	12/12/12	0.0013/0.0013/0.0018	0/0/0
CAD A, 12 h	226	0/12	–	12/12/12	0.0036/0.0037/0.0051	0/0/0
CAD B, 1 h	231	2/12	24.9	12/12/10	0.0004/0.0004/0.0005	0/0/15
CAD B, 12 h	231	3/12	66.2	12/12/11	0.0004/0.0004/0.0006	0/0/1
CAD C, 1 h	168	0/12	–	12/12/12	0.0002/0.0002/0.0003	0/0/0
CAD C, 12 h	168	1/12	43.5	12/12/12	0.0003/0.0003/0.0005	0/0/0

TABLE V

REPEATED CORE QUERIES: FIVE FIXED INSTANCES PER ROW, TEN REPEATS PER METHOD. F/R = FULLCP/BARRCP. CLOSED COUNTS ARE OUT OF 50 CALLS; RANGES GIVE PER-INSTANCE COUNTS OUT OF TEN. TIME IS MEDIAN [Q1,Q3] OF ALL QUERY TOTALS, INCLUDING INCOMPLETE CALLS, NOT UNCENSORED TIME-TO-OPTIMUM. REL. IQR IS THE MEDIAN ACROSS INSTANCES OF EACH INSTANCE'S (Q3–Q1)/MEDIAN TIME (%). THESE ARE DESCRIPTIVE REPEATS, NOT 50 INDEPENDENT INSTANCES.

q	n	Closed F/R	Per-case F/R	Query time F / R (s)	Rel. IQR F/R	Max gap F/R
64	1024	50/50	10–10/10–10	0.453 [0.311,0.604] / 0.216 [0.211,0.238]	2.4/1.9	0/0
64	2048	50/50	10–10/10–10	0.615 [0.545,0.783] / 0.218 [0.214,0.242]	1.5/4.0	0/0
128	1024	18/40	0–9/0–10	5.002 [4.393,5.002] / 3.517 [3.018,4.073]	0.0/2.6	17/17
128	2048	15/39	0–9/0–10	5.004 [4.945,5.004] / 3.548 [3.006,3.922]	0.0/1.3	20/17

VII. CONCLUSION

We derived a budget-anchored region reduction for binary responsibility repair under localized unary changes. Union reversion yields one additive old-gap bound, positive handoff costs define a safe weighted region, and boundary absorption transfers solver bounds. Matched experiments and a weighted-region ablation show benefits on modular and grid structures, while high-connectivity and inexact-plan cases expose clear limits. The result provides a certified preprocessing mechanism within its stated model, motivating tighter certificates and broader feasible sets.

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